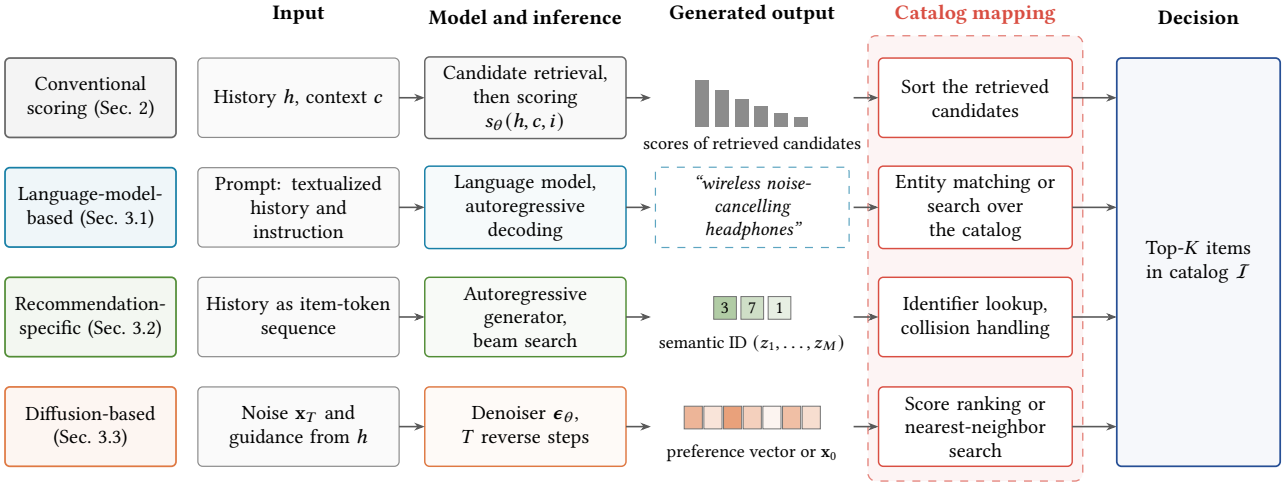

Input
Model and inference
Generated output
Catalog mapping
Decision
Conventional scoring (Sec. 2)
History h, context c
Candidate retrieval, then scoring $s_\theta(h, c, i)$
scores of retrieved candidates
Sort the retrieved candidates
Language-model-based (Sec. 3.1)
Prompt: textualized history and instruction
Language model, autoregressive decoding
"wireless noise-cancelling headphones"
Entity matching or search over the catalog
Recommendation-specific (Sec. 3.2)
History as item-token sequence
Autoregressive generator, beam search
3 7 1
semantic ID $(z_1, \ldots, z_M)$
Identifier lookup, collision handling
Diffusion-based (Sec. 3.3)
Noise $\mathbf{x}_T$ and guidance from h
Denoiser ϵ_θ, T reverse steps
preference vector or $\mathbf{x}_0$
Score ranking or nearest-neighbor search
Top-K items in catalog $\mathcal{I}$